

FULL SDLC · AI-DRIVEN

Optimal AI Agent Architecture

Build → Bug Fix → Feature Enhancement → Testing → Maintenance ·
Minimum tokens, maximum autonomy

 ARCHITECTURE

 AGENTS

 SDLC
FLOW

 OPTION
MATRIX

 TOKEN STRATEGY

 SETUP GUIDE

// CORE AI ORCHESTRATION ARCHITECTURE

The Core Idea: Tiered Agent Orchestration

One **Orchestrator** (Copilot Agent Auto) reads your spec and delegates to **Specialist Agents** per SDLC phase. Each specialist has a constrained context = fewer tokens. Agents hand off via shared memory files, not chat history. Bugs, features, tests, and maintenance all route through the same pipeline — just different specialist prompts.

// ORCHESTRATION LAYERS

LAYER 01

Orchestrator

Copilot Agent Auto @workspace

.github/copilot-instructions.md

Master controller. Reads project spec, decomposes tasks, delegates to sub-agents via structured prompt

Agent

files. Never writes code directly — only plans, validates, and routes. Maintains `_ai/state.json` with current task queue.

~5% of total tokens

Backend Agent Frontend Agent Test Agent

Bug Fix Agent

LAYER 02

Domain Specialists

Each specialist is invoked with a scoped context — only the files relevant to its domain. Backend agent never sees React files. Test agent only sees changed files. This is the #1 token saver.

~60% of total tokens

Lint Gate Build Gate Test Gate

Review Agent

LAYER 03

Validation Layer

Automated gates between every agent handoff. If build fails → Bug Fix Agent is auto-triggered with only the error + changed file. No human needed. Runs via VS Code tasks + GitHub Actions.

~15% of total tokens

`_ai/state.json` `_ai/decisions.md`

`_ai/bug-log.md` `copilot-instructions.md`

LAYER 04

Memory & Context

Persistent memory layer. Agents read/write structured files instead of relying on chat context. This means you can close VS Code and resume — agents pick up exactly where they left off. Decisions are logged so agents don't re-ask the same questions.

~20% of total tokens

BUILD

Backend Agent + Frontend Agent generate full stack from spec + Excel fields. Auto-scaffold, auto-wire, auto-configure.

BUG FIX

Bug Fix Agent receives error + stack trace + relevant file only. Fixes inline. Logs decision to `_ai/bug-log.md` so same bug never recurs.

FEATURE

Feature Agent reads `_ai/features.md`. Adds to existing code with zero breakage — it reviews existing patterns first, then extends.

TESTING

Test Agent runs after every commit. Generates unit + integration tests for changed files only. Not full suite — targeted, cheap.

MAINTENANCE

Weekly Maintenance Agent scans for deprecated deps, N+1 queries, unused code. Produces a diff — you review, one-click apply.

REVIEW

Review Agent checks every PR diff for security issues, missing validations, and API contract breaks. Runs on GitHub Actions.